# Matthew Hutchinson

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UCLA M.S. in Mechanical Engineering with hands-on experience building robotics software and evaluating machine learning systems. Projects include ROS 2/OpenCV autonomy, LLM evaluation, and spatial-data tools, with a focus on testing and system integration.

## EDUCATION

**UCLA | M.S. Mechanical Engineering, Robotics & Artificial Intelligence** Sep 2025 - Aug 2026

Coursework: Deep Learning, Computer Vision, Reinforcement Learning, Natural Language Processing, Soft Robotics, Data Science, Bionic Systems, Autonomous Aerospace Systems

**UCLA | B.S. Mechanical Engineering** Sep 2022 - Jun 2024

Coursework: Robotics (Kinematics, Dynamics & Control), Data Structures & Algorithms, Python, C++, Rapid Prototyping, MATLAB

## TECHNICAL SKILLS

Python, C++, PyTorch, OpenCV, ROS 2, MATLAB, Git, Docker, Tailscale, GCP, SolidWorks, Linux, embedded system, reinforcement learning, control systems, and model evaluation.

## SELECTED PROJECTS

**Knowledge Editing of Large Language Models | UCLA Natural Language Processing**

- Ran reproducible ROME, MEMIT, and IKE experiments on GPT-2 XL using EasyEdit and a Google Cloud T4 GPU, evaluating factual edits on CounterFact and related benchmarks.
- Built 225 diagnostic probes for negation, inverse relations, composition, contradictions, and multi-step reasoning; each method passed fewer than 43%, despite strong direct rewrite scores.
- Tested ROME edits against conflicting retrieved context; documented experiment scripts, results, and reproduction steps in the final report.

**MapLaser | Geospatial Modeling and Digital Fabrication**

- Built a Python pipeline converting OpenStreetMap and shoreline data into scaled, layered laser-cut SVGs with configurable crops, rotation, and road widths
- Developed an interactive web app for worldwide location selection and 3D terrain generation, supporting buildings, GPX overlays, and STL/3MF exports.

**Autonomous Drone Racing | UCLA Deep Learning Project**

- Explored imitation and reinforcement learning for autonomous drone racing in simulation.
- Evaluated perception and state features, trajectory following, and control behavior.

## EXPERIENCE

**UCLA Robotics and Mechanisms Laboratory | Research Intern** Jun 2026 - Present

- Developed a Python kinematic model and motion planner for a Tripteron platform, mapping XYZ targets to three carriage groups and checking joint, rail, and modeled collision constraints.
- Built a browser workspace and path-preview interface with coordinated Klipper execution; analyzed paired-platform motion to compare operating depth and light-span tradeoffs.
- Built a Python/ROS 2 control and diagnostics stack for an 18-actuator hexapod, integrating dual CAN buses, calibration, telemetry logging, fault handling, and synchronized 100 Hz motion.
- Developed and tested coordinated leg motion and walking gaits using motor feedback and bounded trajectories.

**UCLA DevX | Autonomous Rover Developer** Sep 2022 - Jun 2024

- Owned ROS 2/OpenCV autonomy on Raspberry Pi: line following, lost-line recovery, and intersection handling, integrated with Arduino wheel-velocity control.
- Implemented 30 Hz PID wheel control; tested step responses and mitigated camera latency, though course completion remained inconsistent.

**UCLA Engineering MakerSpace | Digital Fabrication Artist** Apr 2024 - Jul 2024

- Mentored 30+ students in prototyping and fabrication; led Illustrator and Fusion 360 workshops and authored equipment quick-reference guides.